CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (ME)

May 2017

ME-309 Design of Machine Elements-I

Date: 30.05.2017, Tuesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Use of PSG design databook is permitted in examination, if required.

SECTION – I

Q - 1 (a) Give answer of following MCQ's. **[07]**

1. Preferred numbers are arranged in
   a. Arithmetic series b. Hyperbolic series
   c. Geometric series d. Random number
2. In the R5 series as given below, the missing number is: 1, 1.60, 2.50, 4.00, ….
   a. 4.90 b. 5.50
   c. 6.00 d. 6.30
3. When the frictional forces helps to apply the brake, then the brake is said to be
   a. Self-energizing brake b. Self-locking brake
4. The groove angle of the pulley for V-belt drive is usually
   a. $20^o – 25^o$ b. $25^o – 32^o$
   c. $32^o – 38^o$ d. $38^o – 45^o$
5. For maximum power, the velocity of the belt will be
   a. $\sqrt{\frac{T}{m}}$ b. $\sqrt{\frac{T}{2m}}$ c. $\sqrt{\frac{T}{3m}}$
6. The cone clutches have become obsolute because of
   a. Small cone angle b. Exposure to dirt and dust
   c. Difficulty in disengaging d. All of these
7. The power transmitted by means of a belt depends upon
   a. Velocity of the belt b. Tension under which the belt is placed on the pulleys
   c. Arc of contact between the belt and the smaller pulley d. All of the above

(b) Derive the frictional torque equation for Disc clutch by considering uniform pressure condition and uniform wear condition. **[07]**

Q – 2 (a) A plate clutch having a single driving plate with contact surface on each side is required to transmit 110 KW at 1250 r.p.m. The outer diameter of the contact surfaces is to be 300 mm. The coefficient of friction is 0.4. **[06]**

(a) Assuming a uniform pressure of .17 N/mm^2; determine the inner diameter of the friction surfaces.

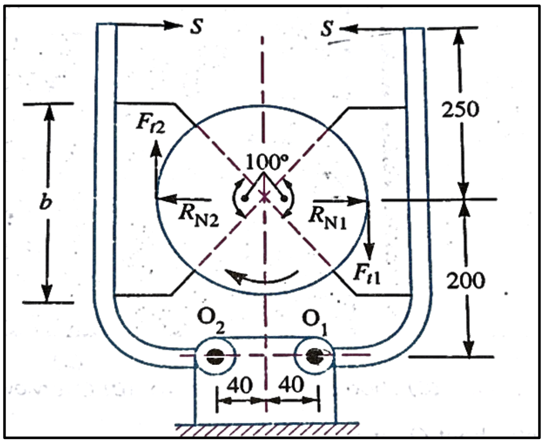
(b) Assuming the same dimensions and the same axial thrust, determine maximum intensity of pressure when uniform wear condition have been reached.

(b) Explain the desired characteristics material should have that use as brake lining. **[06]**

<u>OR</u>

(b) Point out the advantages and disadvantages of V-belt drive over flat belt drive. **[06]**

Q - 3 A double shoe brake, as shown in figure is capable of absorbing a torque of 1400 N-m. The diameter of the brake drum is 350 mm and the angle of contact for each shoe is 100°. If the coefficient of friction between the brake drum and lining is 0.4; find: 1. The spring force necessary to set the brake; and 2: the width of the brake shoes, if the bearing pressure on the lining material is not to exceed 0.3 N/mm^2. **[09]**



<u>OR</u>

An electric motor drives an exhaust fan. Following data are provided: **[09]**

	Motor pulley	Fan pulley
Diameter	400 mm	1600 mm
Angle of warp	2.5 radians	3.78 radians
Coefficient of friction	0.3	0.25
Speed	700 r.p.m.	----
Power transmitted	22.5 KW	----

Calculate the width of 5 mm thick lather flat belt having density of 1000 kg/m^3. Take permissible stress for the belt materials as 2.3 MPa.

SECTION – II

Q - 4 Give answer of following MCQ's. **[07]**

1. A bolt of diameter 33 mm, having safe tensile stress 100 N/mm^2 can safely carry a load of
   - a. 80 kN
   - b. 100 kN
   - c. 120 kN
   - d. 150 kN
2. The criterion of failure for machine parts subjected to fluctuating stress is
   - a. Ultimate tensile strength
   - b. Yield strength
   - c. Endurance limit
   - d. Permissible
3. The efficiency of overhauling screw is
   - a. Equal to 50%
   - b. More than 50%
   - c. Less than 50%
   - d. None of the above
4. The angle of twist for a transmission shaft is inversely propotional to
   - a. Shaft diameter
   - b. (Shaft diameter)^2
   - c. (Shaft diameter)^3
   - d. (Shaft diameter)^4
5. The compressive stress induced in a square is
   - a. Equal to shear stress
   - b. Four times of shear stress
   - c. Twice of shear stress
   - d. Half of shear stress
6. A cup is provided in screw jack
   - a. To reduce the friction
   - b. To prevent rotation of load
   - c. To increase load capacity
   - d. To increase efficiency
7. For distortion energy theory, the shape of the region of safety on σ_1 and σ_2 co-ordinate system is
   - a. Square
   - b. Hexagon
   - c. Ellipse
   - d. circle

Q - 5 (a) Derive an equation for thick cylinder with both ends closed and made of ducitle material and According to distortion energy theory of failure, using all three principal stresses, prove that the wall thickness is given by, **[08]**

$$t = \frac{Di}{2}\left[\sqrt{\frac{\sigma}{(\sigma - \sqrt{3}\,Pi)}} - 1\right]$$

(b) What is stress concentration? Explain the techniques to reduce stress concentration. **[06]**

OR

(b) What is the difference between gerber curve and Soderberg and goodman line? **[06]**

Q – 6 Solve any two. **[14]**

(a) Design a cast iron flange coupling for a mild steel shaft transmitting 90 kW at 250 rpm. The allowable shear stress in the shaft is 40 MPa and the angle of twist is not to exceed 1^o in a length of 20 diameters. The allowable shear stress in the coupling bolts is 30 MPa. Take width and thickness of key is 25 mm and 14 mm respectively.

(b) A screw press is to exert a force of 40 kN. The unsupported length of the screw is 400 mm. Nominal diameter of screw is 50 mm. The screw has square threads with pitch equal to 10 mm. The material of the screw and nut are medium carbon steel and cast

iron respectively. For the steel used take ultimate crushing stress as 320 MPa, yield stress in tension or compression as 200 MPa and that in shear as 120 MPa. Allowable shear stress for cast iron is 20 MPa and allowable bearing pressure between screw and nut is 12 N/mm2. Young's modulus for steel = 210 kN/mm2.
Determine the factor of safety of screw against failure. Find the dimensions of the nut. What is the efficiency of the arrangement? Take coefficient of friction between steel and cast iron as 0.13, C = 0.25 and assume that load is uniformly distributed over the threads in contact.

(c) Determine the life cycles of the mechanical component subjected to complete reversed bending stresses as follows:

1. 350 N/mm^2 for 85% of time
2. 500 N/mm^2 for 3% of time
3. 400 N/mm^2 for remaining time

The component is made of plain carbon steel 50C4 having S_{ut} = 660 N/mm^2 and endurance limit of component is 280 $N/mm^{2.}$

****